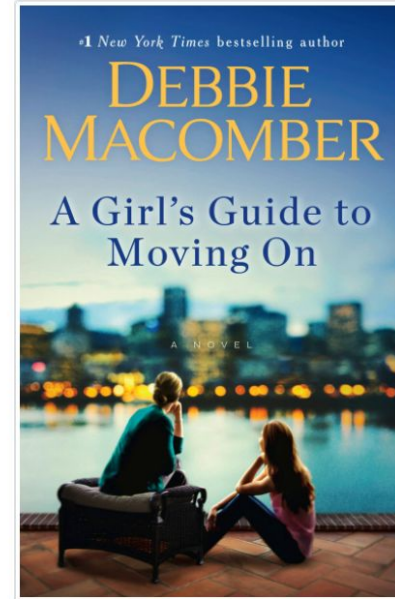
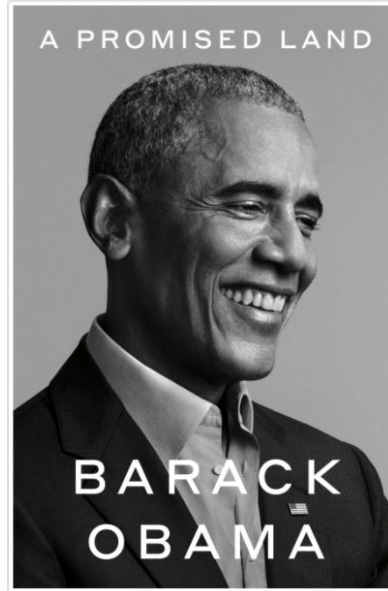
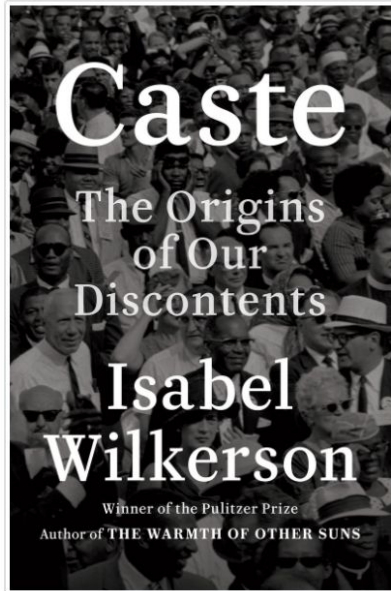

COMS 4170 · SPRING 2021

Flask Template Starter Kit

Prof. Celeste Layne
Monday February 22, 2021





JS books.js

<> books.html

server.py X

```

1  # import os module
2  import os
3
4  # import flask
5  from flask import Flask, url_for
6
7  # import template engine
8  from flask import render_template
9
10 from flask import json
11
12 # create an app instance
13 app = Flask(__name__)
14
15 # import data
16 filename = os.path.join(app.static_folder, 'data.json')
17 with open(filename) as books_data:
18     books = json.load(books_data)
19
20 # show all books route
21 @app.route('/books')
22
23 # write a method to show all books
24 def show_all_books():
25     global books
26
27     return render_template('books.html', books=books)

```

The os module provides a way to interact with the file system.

Import data from external JSON file using os.path module

Data is now accessible in the server to be rendered to the client.

JS books.js

data.json X

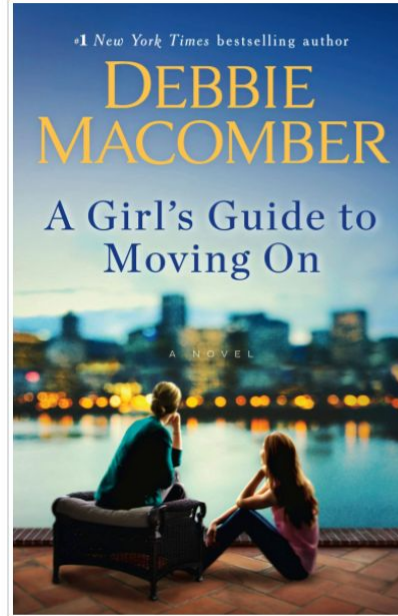
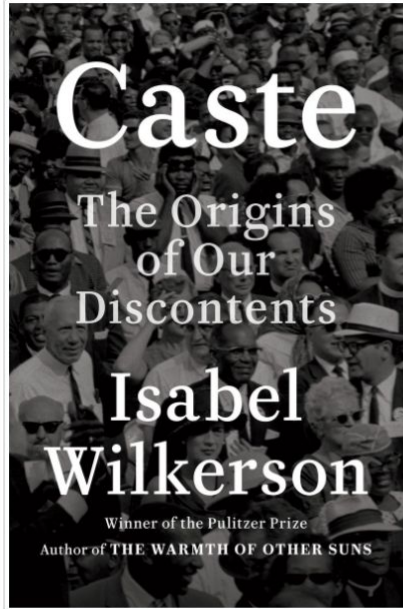
<> books.html

```

1  {
2    "status": "OK",
3    "copyright": "Copyright (c) 2019 The New York Times Company. All Rights Reserved.",
4    "num_results": 1,
5    "last_modified": "2016-03-11T13:09:01-05:00",
6    "results": [
7      {
8        "list_name": "Hardcover Fiction",
9        "display_name": "Hardcover Fiction",
10       "bestsellers_date": "2016-03-05",
11       "published_date": "2016-03-20",
12       "rank": 5,
13       "rank_last_week": 2,
14       "weeks_on_list": 2,
15       "asterisk": 0,
16       "dagger": 0,
17       "amazon_product_url": "http://www.amazon.com/Girls-Guide-Moving-On-Novel-ebook/dp/B00ZNE17B4?tag=thenewyorktim-20",
18       "isbns": [
19         {
20           "isbn10": "0553391925",
21           "isbn13": "9780553391923"
22         }
23       ],
24       "book_details": [
25         {
26           "title": "A GIRL'S GUIDE TO MOVING ON",
27           "description": "A mother and her daughter-in-law both leave unhappy marriages and take up with new men.",
28           "contributor": "by Debbie Macomber",
29           "author": "Debbie Macomber",
30           "contributor_note": "",
31           "price": 0,
32           "image": "https://images-na.ssl-images-amazon.com/images/I/81Bt6Vl5bJL.jpg",
33           "age_group": "",
34           "publisher": "Ballantine",
35           "primary_isbn13": "9780553391923",
36           "primary_isbn10": "0553391925"
37         }
38       ],
39       "reviews": [
40         {
41           "book_review_link": "",
42           "first_chapter_link": "",
43           "sunday_review_link": "",
44           "article_chapter_link": ""
45         }
46       ]
47     }
48   ]
49 }

```

IMPORT DATA



Elements Sources Network Console »

top Filter Default levels

```
books:48
ready!
books.js:2
books.js:16

▼ Object
  amazon_product_url: "http://www.amazon.com/Girls-Guide-Moving-0..."
  asterisk: 0
  bestsellers_date: "2016-03-05"
  ▶ book_details: [{...}]
  dagger: 0
  display_name: "Hardcover Fiction"
  ▶ isbns: [{...}]
  list_name: "Hardcover Fiction"
  published_date: "2016-03-20"
  rank: 5
  rank_last_week: 2
  ▶ reviews: [{...}]
  weeks_on_list: 2
  ▶ __proto__: Object
books.js:16

▼ Object
  amazon_product_url: "https://www.amazon.com/dp/1524763160?tag=N..."
  asterisk: 0
  bestsellers_date: "2021-02-13"
  ▶ book_details: [{...}]
  dagger: 0
  display_name: "Hardcover Nonfiction"
  ▶ isbns: (4) [{...}, {...}, {...}, {...}]
  list_name: "Hardcover Nonfiction"
  published_date: "2021-02-28"
  rank: 3
  rank_last_week: 4
  ▶ reviews: [{...}]
  weeks_on_list: 13
  ▶ __proto__: Object
books.js:16

▼ Object
  amazon_product_url: "https://www.amazon.com/dp/0593230256?tag=N..."
  asterisk: 0
  bestsellers_date: "2021-02-13"
  ▶ book_details: [{...}]
  dagger: 0
  display_name: "Hardcover Nonfiction"
  ▶ isbns: (3) [{...}, {...}, {...}]
  list_name: "Hardcover Nonfiction"
  published_date: "2021-02-28"
  rank: 8
  rank_last_week: 7
  ▶ reviews: [{...}]
  weeks_on_list: 28
  ▶ __proto__: Object
```

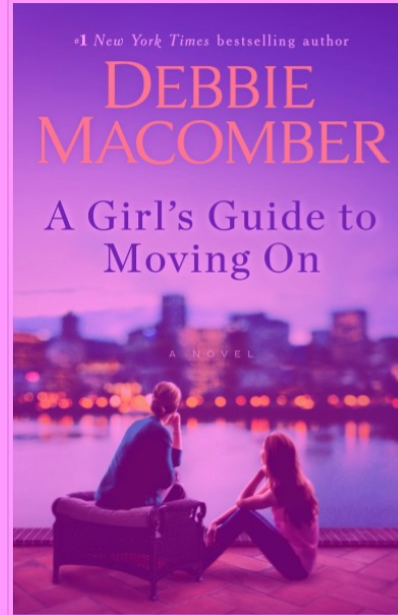
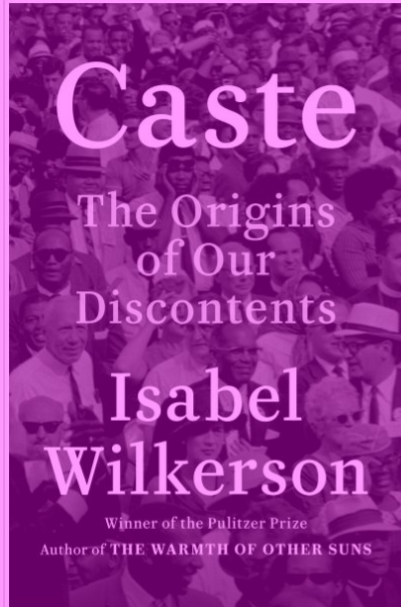
MOCKUP WITH DATA

New York Times Book List

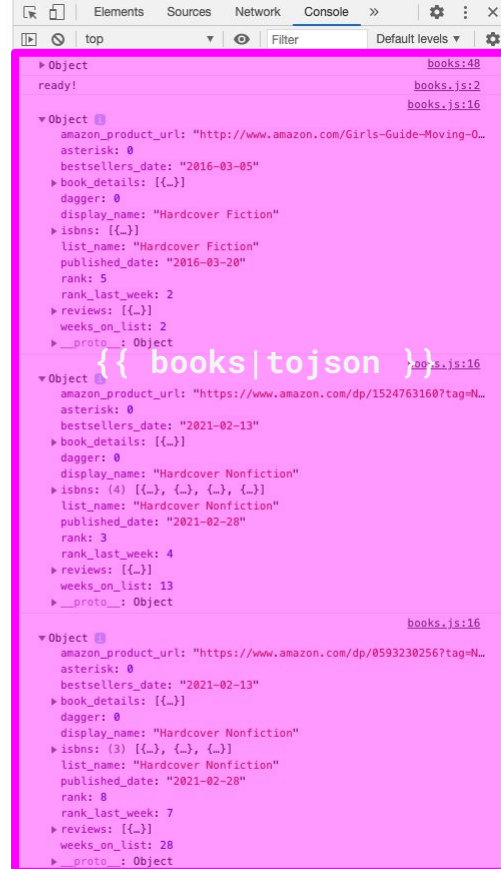
{% block navigation%}{% endblock navigation%}

Search

Search



{% block content%}{% endblock content%}



MOCKUP WITH DATA

Template Inheritance

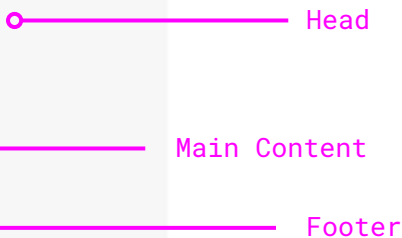
The most powerful part of Jinja is template inheritance. Template inheritance allows you to build a base “skeleton” template that contains all the common elements of your site and defines **blocks** that child templates can override.

Sounds complicated but is very basic. It’s easiest to understand it by starting with an example.

Base Template

This template, which we’ll call `layout.html`, defines a simple HTML skeleton document that you might use for a simple two-column page. It’s the job of “child” templates to fill the empty blocks with content:

```
<!doctype html>
<html>
  <head>
    {% block head %}
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
    <title>{% block title %}{% endblock %} - My Webpage</title>
    {% endblock %}
  </head>
  <body>
    <div id="content">{% block content %}{% endblock %}</div>
    <div id="footer">
      {% block footer %}
      &copy; Copyright 2010 by <a href="http://domain.invalid/">you</a>.
      {% endblock %}
    </div>
  </body>
</html>
```



Head

Main Content

Footer

In this example, the `{% block %}` tags define four blocks that child templates can fill in. All the `block` tag does is tell the template engine that a child template may override those portions of the template.

```
JS books.js  <> books.html  <> layout.html X  ...
1  <!doctype html>
2  <html lang="en">
3  <head>
4    <!-- Required meta tags -->
5    <meta charset="utf-8">
6    <meta name="viewport" content="width=device-width, initial-scale=1">
7
8    <!-- Bootstrap CSS -->
9    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.0-beta2/dist/css/bootstrap.min.
      css" rel="stylesheet" integrity="sha384-BmbxuPwQa2lc/FVzBcNJ7UAyJxM6wuuqIj61tLrc4wSX0szH/
      Ev+nYRRuWlolfffl" crossorigin="anonymous">
10
11    <title>New York Times Book List</title>
12  </head>
13  <body>
14    <div class="container">
15      <div class="row">
16        {% block navigation %}{% endblock %}
17      </div>
18      <div class="row">
19        {% block content %}{% endblock %}
20      </div>
21    </div>
22
23    <!-- jQuery CDN -->
24    <script src="https://code.jquery.com/jquery-1.12.4.js"></script>
25
26    <!-- Bootstrap CDN files -->
27    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.0-beta1/dist/js/bootstrap.min.
      js" integrity="sha384-pQQkAEnwaBkjpqZ8RU1ff1AKtTcHJwF13pblpTLUXybjjHpmYo79HY3hIi4NKxyj"
      crossorigin="anonymous"></script>
28
29    <!-- Custom JavaScript -->
30    <script src="{% url_for('static', filename='books.js') %}"></script>
31
32    <!-- Model data from server.py -->
33    <script>
34      const books = JSON.parse('{{ books|tojson }}')
35      console.log(books)
36    </script>
37
38  </body>
39  </html>
```

Navigation Bar

Main Content (books
get rendered here)

BASE TEMPLATE

Child Template

Child Template

A child template might look like this:

```
{% extends "layout.html" %}
{% block title %}Index{% endblock %}
{% block head %}  Head
    {{ super() }}
    <style type="text/css">
        .important { color: #336699; }
    </style>
{% endblock %}
{% block content %}  Main Content
    <h1>Index</h1>
    <p class="important">
        Welcome on my awesome homepage.
    </p>
{% endblock %}
```

The `{% extends %}` tag is the key here. It tells the template engine that this template “extends” another template. When the template system evaluates this template, first it locates the parent. The extends tag must be the first tag in the template. To render the contents of a block defined in the parent template, use `{{ super() }}`.

```
JS books.js    books.html X    layout.html
1  {% extends 'layout.html' %}
2
3  {% block navigation %}
4  <nav class="navbar navbar-dark bg-dark">
5    <div class="container-fluid">
6      <a class="navbar-brand">New York Times Book List</a>
7      <form class="d-flex">
8        <input class="form-control me-2" type="search" placeholder="Search"
9          aria-label="Search">
10       <button class="btn btn-outline-light" type="submit">Search</button>
11     </form>
12   </div>
13 </nav>
14 {% endblock navigation %}
15
16 {% block content %}
17   <!-- render list of books -->
18   <div class="row books"></div>
19 {% endblock content %}
```

Reference to base
template

Code for the
navigation bar

Code for the main
content (i.e. list of
books)

CHILD TEMPLATE